

About

INSTROLE offers real-time monitoring and record of precise, quantitative measures of one's bipedal weight distribution during quiet standing and dynamic gait for young and elderly with and without balance disorders.

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IIT Gandhinagar



Sarita G Gelot Laboratory for
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Affective Computing Systems



The Next Generation

INSTRUMENTED SMART SHOE INSOLE



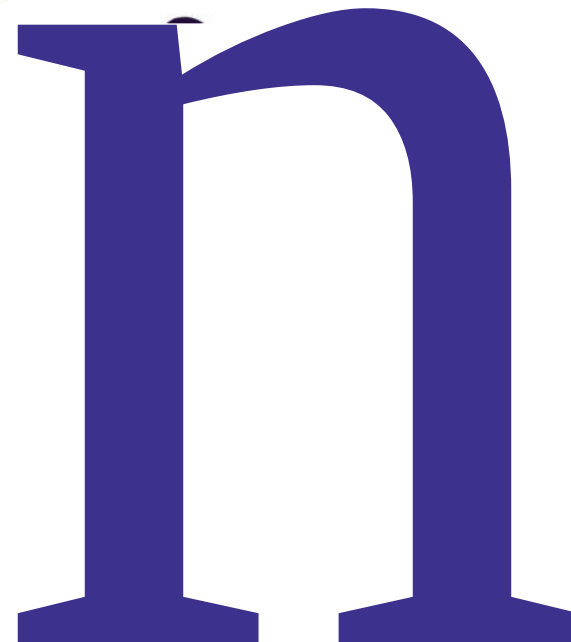
A product by Oreejoy



Partner for
*healthy balance
and gait*



INSTROLE
BiPeD



How to use

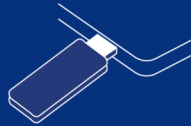
INSTROLE *BiPeD* ?



Insert the
INSTROLE *BiPeD*
into a shoe



Establish and verify
connections from
INSTROLE *BiPeD* to
tablet PC



Follow the INSTROLE *BiPeD*
GUI on a tablet PC. Collect
and store output in form of
gait indices and weight
distribution for quiet
standing and dynamic
gait conditions



Get a consolidated
report for scrutiny by
medical expert



Features

- Sole with Embedded sensors connected to a microcontroller for data acquisition.
- Wireless connectivity of microcontroller with tablet PC.
- Measures and records weight distribution while at rest and in motion assisting in diagnostics such as flat foot.
- Measures and quantifies gait in terms of step time, stride time, cadence, stance and swing phases, etc.
- Suitable for monitoring over any ground surface or treadmill-assisted walk/run.

Benefits

Complement clinical diagnosis by offering objective metrics while saving administration time

Offer visualization of balance and gait conditions to patients to realize the possible risk factors

Monitor patient's progress thereby motivating the patient to undergo treatment

Quantified screening for early diagnosis to prevent possible future risks

Easy quantification of weight distribution with consolidated report

User-friendly system of hardware and software